

Would GenAI-Assisted Screening Deter Older Applicants?

Procedural Fairness and Application Burden in Singapore Public Service Recruitment

Working proposal for advisor discussion · August 2026

Setting	Singapore public-service recruitment; a fictional Careers@Gov-style vacancy; explicitly hypothetical pilot
Core design	Cognitive interviews → 2 × 2 randomized vignette experiment → explanatory interviews
Randomized factors	Screening process (human-led / GenAI-assisted) × procedural safeguards (standard notice / safeguard package)
Primary outcomes	Application willingness; behavioral persistence in opening, starting, and completing an optional mock application step
Primary estimand	Screening × safeguard interaction on a preregistered application-response index
Position in PhD	First qualifying paper; no agency access required; pilot can begin independently

1. Motivation and research question

When public organizations consider AI-assisted recruitment, the usual question is whether a tool can make screening faster or more consistent. An equally important policy question is whether disclosing AI involvement changes who is willing to apply. Older applicants may be especially attentive to whether a system can recognize career transitions, non-linear work histories, and experience accumulated over time. Any deterrent effect would alter the applicant pool before formal selection begins.

The study asks whether disclosure of GenAI-assisted resume screening in an explicitly hypothetical Singapore public-service recruitment scenario reduces application willingness and behavioral persistence; whether a package of stated criteria, a short explanation, independent human reconsideration, and factual-error correction attenuates that effect; and whether the effect varies with age and perceived employability.

2. Research gap and theoretical framework

Keppeler (2024) shows that AI disclosure in public-sector recruitment can reduce applicant interest. A separate hiring literature documents age stereotypes and age differences in reemployment. These literatures rarely meet, and existing evidence says little about whether concrete safeguards alter applicant reactions. The study examines responses to an AI-assisted hiring process; it does not test whether a real screening tool discriminates by age.

- Algorithm aversion explains why machine involvement in evaluation may trigger avoidance or lower trust.
- Selection procedural justice explains why job relatedness, consistent criteria, explanation, voice, and reconsideration shape applicant judgments.
- Age-discrimination and reemployment research motivates tests of age heterogeneity, but age is not randomized and is not given a causal interpretation.
- Administrative burden is used narrowly to describe anticipated psychological and application costs, without treating public employment as equivalent to claiming benefits or using a public service.
- Public accountability: Busuioc (2021) provides an institutional basis for information, explanation and judgment, and consequences within the safeguard package.

Mechanism measures are used to assess whether the results are consistent with the proposed explanation. They are not treated as causally identified mediators. Primary outcomes are measured before procedural-fairness, psychological-cost, and manipulation-check items to reduce priming.

3. Mixed-methods research design

- Phase 1: conduct 12-16 cognitive interviews across planned age bands to test comprehension of the posting, GenAI process, and safeguard language.

- Phase 2: field a 2×2 between-subjects randomized vignette experiment. The screening descriptions are matched in length, task, and HR final authority.
- Phase 3: purposively select 20-25 participants across age, treatment, and response profiles for explanatory interviews.

Factor	Condition A	Condition B
Screening process	Human-led: an HR officer reads the full resume, prepares a competency summary against stated criteria, and makes the shortlisting decision.	GenAI-assisted: a tool reads the full resume, conducts semantic matching, and generates the same type of summary; HR retains the final decision.
Procedural design	Standard notice: states the screening process and how applicants will be notified.	Safeguard package: stated criteria, a short reason if screened out, independent human reconsideration, and a factual-error correction channel.

4. Sample, measurement, and field sequence

The study first fixes one mid-level occupation and its education and experience requirements. It then recruits eligible Singapore citizens and permanent residents aged approximately 30-59, stratified across ages 30-39, 40-49, and 50-59. Continuous age is used for the primary heterogeneity model. Tenure, career stage, prior Careers@Gov use, perceived job fit, and perceived employability are prespecified secondary measures.

- Primary outcome 1: application willingness, measured with a preregistered multi-item index.
- Primary outcome 2: behavioral application persistence, measured by whether a participant opens, starts, and completes an optional mock application step.
- Exploratory outcome: completion time. Because it also reflects typing speed, digital familiarity, and device, it is not labeled directly as compliance cost.
- Explanatory measures: procedural fairness, perceived opacity, anticipated age bias, organizational attractiveness, and psychological application cost.
- Order: primary outcomes first, then explanatory measures and manipulation checks. Manipulation-check failures diagnose the pilot and are not used as post-treatment exclusions in the main intention-to-treat analysis.

5. Hypotheses

- H1: GenAI-assisted screening disclosure reduces application willingness and behavioral persistence relative to the matched human-led process.
- H2: The safeguard package attenuates the negative effect of GenAI-assisted screening.
- H3: The deterrent effect is larger at higher ages and among applicants reporting lower perceived employability. This is treatment-effect heterogeneity.
- H4 (exploratory): lower procedural fairness, greater opacity, anticipated age bias, and psychological application cost are consistent with the treatment pattern; interviews explain divergent responses.

6. Analysis and sample size

The primary confirmatory estimand is the screening $\times$ safeguard interaction on a preregistered application-response index; both treatment main effects are also reported. Age heterogeneity is estimated with continuous age and prespecified interactions and is not interpreted as a causal effect of age. Secondary outcomes are grouped into theory-based families with multiplicity adjustment. Qualitative analysis combines a prespecified framework with inductive coding across treatment and response profiles.

The planning range for the full experiment is approximately 900-1,200 respondents. The final sample is preregistered after simulation-based power analysis using a smallest effect of substantive interest, expected cell balance, measure reliability, and the primary interaction. A 120-160 person pilot checks comprehension, realism, the distinctiveness of GenAI from conventional algorithms, and outcome variance. Its unstable interaction estimate is not treated as the true effect size.

7. Identification and evidence boundaries

- Random assignment identifies effects of the stated screening and safeguard conditions on outcomes inside the study, not actual job-application behavior.
- The study does not establish the accuracy or age bias of a real system and does not describe current Singapore GenAI recruitment policy.
- Age, career stage, and perceived employability are measured variables; their interactions show heterogeneity, not causal effects of those characteristics.
- The safeguard condition is a package, so individual components are not separately identified.
- A human-versus-GenAI comparison cannot isolate a uniquely generative-AI response from automation reactions in general. If pilot participants cannot distinguish the process, the full study should revise the manipulation or add a conventional-AI arm.

8. Contribution, feasibility, and next steps

The study connects applicant reactions, procedural design, and age heterogeneity in one Singapore public-recruitment experiment. Theoretically, it tests whether selection procedural justice helps explain and attenuate AI disclosure effects, while assessing the usefulness of application-cost measures in a state-as-employer setting. Practically, it evaluates a safeguard package before any real pilot is scaled.

The project does not require agency access and is suitable as the first qualifying paper. The main feasibility risks are insufficient recruitment in older age bands, imperfect functional matching between screening descriptions, and a mock behavior task that lacks realism. The next step is to select one occupation, draft all four experimental versions, conduct 12-16 cognitive interviews, and use prespecified criteria to decide whether to proceed to the 120-160 person pilot.

References

- Busuioc, M. (2021). Accountable artificial intelligence: Holding algorithms to account. *Public Administration Review*, 81(5), 825-836. <https://doi.org/10.1111/puar.13293>
- Dietvorst, B. J., Simmons, J. P., & Massey, C. (2015). Algorithm aversion: People erroneously avoid algorithms after seeing them err. *Journal of Experimental Psychology: General*, 144(1), 114-126. <https://doi.org/10.1037/xge0000033>
- Gilliland, S. W. (1993). The perceived fairness of selection systems: An organizational justice perspective. *Academy of Management Review*, 18(4), 694-734. <https://doi.org/10.5465/amr.1993.9402210155>
- Halling, A., & Baekgaard, M. (2024). Administrative burden in citizen-state interactions: A systematic literature review. *Journal of Public Administration Research and Theory*, 34(2), 180-195. <https://doi.org/10.1093/jopart/muad023>
- Hausknecht, J. P., Day, D. V., & Thomas, S. C. (2004). Applicant reactions to selection procedures: An updated model and meta-analysis. *Personnel Psychology*, 57(3), 639-683. <https://doi.org/10.1111/j.1744-6570.2004.00003.x>
- Keppeler, F. (2024). No thanks, dear AI! Understanding the effects of disclosure and deployment of artificial intelligence in public sector recruitment. *Journal of Public Administration Research and Theory*, 34(1), 39-52. <https://doi.org/10.1093/jopart/muad009>
- Moynihan, D., Herd, P., & Harvey, H. (2015). Administrative burden: Learning, psychological, and compliance costs in citizen-state interactions. *Journal of Public Administration Research and Theory*, 25(1), 43-69. <https://doi.org/10.1093/jopart/muu009>
- Neumark, D., Burn, I., & Button, P. (2019). Is it harder for older workers to find jobs? New and improved evidence from a field experiment. *Journal of Political Economy*, 127(2), 922-970. <https://doi.org/10.1086/701029>
- Wanberg, C. R., Kanfer, R., Hamann, D. J., & Zhang, Z. (2016). Age and reemployment success after job loss: An integrative model and meta-analysis. *Psychological Bulletin*, 142(4), 400-426. <https://doi.org/10.1037/bul0000019>
- VITAL. Recruitment services and AI/ML recruitment pilots. <https://www.vital.gov.sg/services/recruitment/>
- Careers@Gov. Singapore Public Service job portal. <https://jobs.careers.gov.sg/>